

# Derivation of Fair Coupon Rate for Autocallable Structured Product

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## 1 High-Level Summary of the Setup & Parameters

We perform a Monte Carlo simulation to find the fair coupon rate for an autocallable structured product (an autocall) tied to a basket of 5 international equity indices/ETFs (3119.HK, EWY, EWT, 2644.T, 00878.TW)[cite: 1]. It accounts for a multi-year term structure of interest rates, correlated asset returns, annual autocall barriers, and a downside barrier protection feature at maturity[cite: 1].

1. **High-Level Summary of the Setup & Parameters:** Defines simulation size, asset weights, annual risk-free rates, volatility matrices, and asset cross-correlations[cite: 1].
2. **Direct Valuation Benchmark:** Uses supplied cumulative distribution function (CDF) probabilities to compute theoretical prices as a sanity check[cite: 1].
3. **Monte Carlo Path Generation:** Generates correlated geometric Brownian motion paths for all 5 assets across 5 annual steps using Cholesky decomposition[cite: 1].
4. **Autocall Logic:** Evaluates whether the basket performance exceeds the barrier at each year[cite: 1]. If triggered, the product "autocalls" and redeems early[cite: 1].
5. **Maturity & Payoff:** For paths that survive all 5 years without autocalling, a conditional downside protection payout is applied based on the final basket level[cite: 1].
6. **Coupon Pricing:** Solves for the breakeven coupon yield for the cash-flow convention and estimates standard errors using the delta method[cite: 1].

## 2 Mathematical Formulas

### 2.1 A. Term Structure & Discount Factors

With a discrete rate vector for year, the continuous rate equivalent (when `CONTINUOUS_RATE = False`) is computed via log-compounding[cite: 1]:

$$r_j = \ln(1 + R_j) \quad (1)$$

The discount factor for cash flows at time  $t_j$  is[cite: 1]:

$$D_j = e^{-r_j t_j} \quad (2)$$

### 2.2 B. Asset Price Simulation (Geometric Brownian Motion)

The asset price paths are simulated across annual intervals  $\Delta t = 1$ [cite: 1]. The multi-asset diffusion step for asset  $i$  at interval  $j$  is[cite: 1]:

$$X_{i,j} = X_{i,j-1} \exp \left( \left( r_j - q_{i,j} - \frac{1}{2} \Delta V_{i,j} \right) + \sqrt{\Delta V_{i,j}} Z_{i,j} \right) \quad (3)$$

Where  $Z_{i,j}$  are correlated standard normal random variables generated via[cite: 1]:

$$Z = \text{RNG} \times L^T \quad (4)$$

( $L$  is the Cholesky factor of the correlation matrix  $\rho$ , such that  $\rho = LL^T$ )[cite: 1].  $\Delta V_{i,j}$  represents the interval variance for asset  $i$  in year  $j$ [cite: 1].

### 2.3 C. Basket Performance & Autocall Trigger

The basket value  $B_j$  at year  $j$  is a weighted sum of the normalized asset prices[cite: 2]:

$$B_j = \sum_{i=1}^5 w_i X_{i,j} \quad (5)$$

An autocall occurs at the earliest year  $k$  where[cite: 2]:

$$B_k \geq H_k \quad (\text{Barrier for year } k) \quad (6)$$

### 2.4 D. Maturity Payout (Year 5)

If the product is never autocalled through 5 years (never event), the final payoff  $R_5$  is determined by the final basket value  $B_5$ [cite: 2]:

$$R_5 = \begin{cases} 1.0 & \text{if } B_5 \geq 0.70 \\ B_5 & \text{if } B_5 < 0.70 \end{cases} \quad (7)$$

### 2.5 E. Coupon Yield Calculation ( $y$ )

The fair coupon  $y$  solves the present value equation where the expected present value of inflows equals the initial investment (normalized to 1.0)[cite: 2]:

$$1.0 = E[a + y \cdot b] \quad (8)$$

$$y = \frac{1.0 - E[a]}{E[b]} \quad (9)$$

**Case A ( $b_A$ ):** Coupons accrue only up to the early call or maturity year, omitting never-called paths from coupon payments[cite: 2].

## 3 Pricing Results

The Monte Carlo simulation yields the following valuation and statistical accuracy metrics under Case A:

| Pricing Metric                  | Value                    |
|---------------------------------|--------------------------|
| Fair Coupon                     | 13.752219%               |
| Standard Error (SE)             | 2.236 bp                 |
| Approx. 95% Confidence Interval | [13.708396%, 13.796041%] |

## 4 Grids and Matrices

### 4.1 Term Structure of Interest Rates

### 4.2 Correlation Matrix ( $\rho$ )

Represents the pairwise linear correlations between the 5 underlying assets[cite: 3]:

| Year ( $t$ )     | Risk-Free Rate ( $R$ ) | Continuous Rate ( $r = \ln(1 + R)$ ) | Discount Factor ( $D$ ) |
|------------------|------------------------|--------------------------------------|-------------------------|
| 1                | 4.78%                  | 4.671%                               | 0.9544                  |
| 2                | 5.30%                  | 5.166%                               | 0.9015                  |
| 3                | 5.52%                  | 5.375%                               | 0.8466                  |
| 4 (interpolated) | 5.605%                 | 5.454%                               | 0.7981                  |
| 5                | 5.69%                  | 5.534%                               | 0.7510                  |

| Asset    | 3119.HK | EWY   | EWT   | 2644.T | 00878.TW |
|----------|---------|-------|-------|--------|----------|
| 3119.HK  | 1.000   | 0.534 | 0.741 | 0.863  | 0.562    |
| EWY      | 0.534   | 1.000 | 0.587 | 0.491  | 0.438    |
| EWT      | 0.741   | 0.587 | 1.000 | 0.614  | 0.689    |
| 2644.T   | 0.863   | 0.491 | 0.614 | 1.000  | 0.517    |
| 00878.TW | 0.562   | 0.438 | 0.689 | 0.517  | 1.000    |

### 4.3 Implied Volatility Grid (vol)

Annualized volatility parameters for each asset across the 5 annual periods[cite: 4]:

| Asset / Year | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|--------------|--------|--------|--------|--------|--------|
| 3119.HK      | 30.45% | 32.34% | 33.50% | 36.70% | 37.50% |
| EWY          | 41.50% | 42.70% | 44.50% | 44.50% | 44.50% |
| EWT          | 32.40% | 35.80% | 36.30% | 36.30% | 36.30% |
| 2644.T       | 23.00% | 24.90% | 25.60% | 25.60% | 25.90% |
| 00878.TW     | 7.00%  | 9.00%  | 12.00% | 12.00% | 12.00% |

### 4.4 Interval Variances ( $\Delta V$ )

Calculated incremental variances ( $\sigma^2 \times \Delta t$ ) used to drive the variance diffusion step per year[cite: 4]:

| Asset / Interval | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|------------------|--------|--------|--------|--------|--------|
| 3119.HK          | 0.0927 | 0.1126 | 0.1177 | 0.1633 | 0.1744 |
| EWY              | 0.1722 | 0.1868 | 0.2078 | 0.2078 | 0.2078 |
| EWT              | 0.1050 | 0.1340 | 0.1384 | 0.1384 | 0.1406 |
| 2644.T           | 0.0529 | 0.0620 | 0.0655 | 0.0655 | 0.0689 |
| 00878.TW         | 0.0049 | 0.0081 | 0.0162 | 0.0162 | 0.0162 |